

The Living World



"The living world is full of diverse organisms, each with unique features but sharing some common characteristics of life."

Examples

1. What Defines "Living"?

Living organisms are those that show certain special characteristics that non-living things do not.

The basic characteristics of living organisms are:

- Growth** – Increase in size, mass or number of cells.
- Reproduction** – Ability to produce offspring of their own kind.
- Metabolism** – All the chemical activities that occur in an organism.
- Response to Stimuli** – Ability to detect and respond to changes in the environment.
- Movement** – Movement of whole body or its parts.
- Homeostasis** – Ability to maintain a stable internal environment.
- Adaptation** – Ability to adjust and survive in a particular environment.



Plant grows from a seed into a big tree.



Rabbits reproduce and give birth to young ones.



We eat food which is used in various metabolic activities.



Mimosa plant folds its leaves on touch.



Humans can move from one place to another.



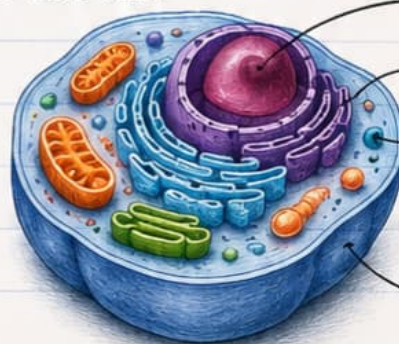
We sweat to maintain body temperature.



Cactus is adapted to survive in deserts with less water.

Building Block of Life – Cell

All living organisms are made up of one or more cells.



- Nucleus** – Control centre of the cell
- Cytoplasm** – Site of various activities
- Cell membrane** – Protects the cell and regulates movement of substances
- Mitochondria** – Powerhouse of the cell

Common Life Processes

- Nutrition** – Obtaining and using food.
- Respiration** – Releasing energy from food.
- Transportation** – Movement of substances inside the body.
- Excretion** – Removal of waste products.
- Reproduction** – Production of new individuals.

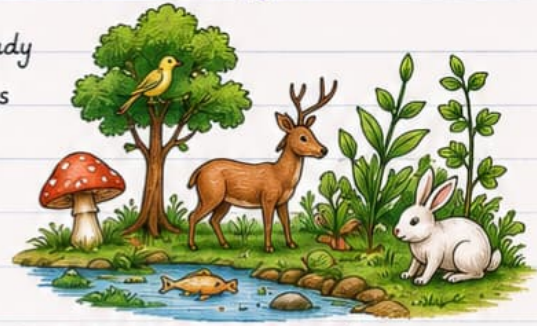
Key Point

All living organisms, whether simple or complex, follow these life processes in different ways.

Thus, life is defined by a set of characteristics and processes that keep organisms alive.

2. Biological Classification and Nomenclature

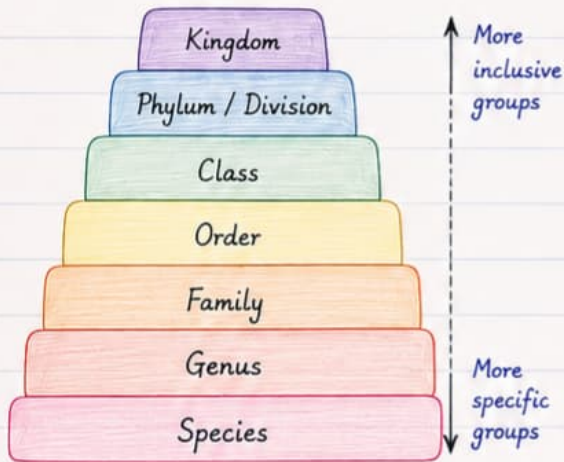
The living world is so diverse that it is difficult to study each organism separately. So, scientists group organisms based on similarities and differences. This process is called **biological classification**.



2.1 Classification

- Grouping of organisms into categories or groups based on their similarities and differences.
- Helps in easy study, identification and understanding of relationships.

2.2 Hierarchy of Classification



2.3 Nomenclature

Nomenclature is the system of giving names to organisms.

Binomial Nomenclature:

- Proposed by **Carolus Linnaeus**.
- Each organism has two names.
- First name → **Genus** (capital letter)
- Second name → **Species** (small letter)
- Written in italic (or underlined separately when handwritten)

Example:

Human → **Homo** sapiens
 ↓ ↓
 Genus Species
 (Capital letter) (Small letter)

2.4 Five Kingdom Classification

Kingdom	Characteristics	Examples
Monera	Unicellular, prokaryotic, no true nucleus.	Bacteria, Cyanobacteria
Protista	Mostly unicellular, eukaryotic.	Amoeba, Paramecium
Fungi	Eukaryotic, non-green, absorptive nutrition.	Mushroom, Yeast
Plantae	Multicellular, eukaryotic, autotrophic.	Moss, Fern, Mango tree
Animalia	Multicellular, eukaryotic, heterotrophic, no cell wall.	Human, Dog, Butterfly

Key Points

- ★ Classification is based on natural affinities.
- ★ The more characters two organisms share, the closer they are in classification.
- ★ Scientific names are universal and accepted globally.
- ★ It avoids confusion caused by local or common names.

Example

- Mango – common name
- *Mangifera indica* – scientific name
- Rice – common name
- *Oryza sativa* – scientific name



In Short

Classification organizes the **diversity** of life into groups, while nomenclature provides each organism a **unique scientific identity**.

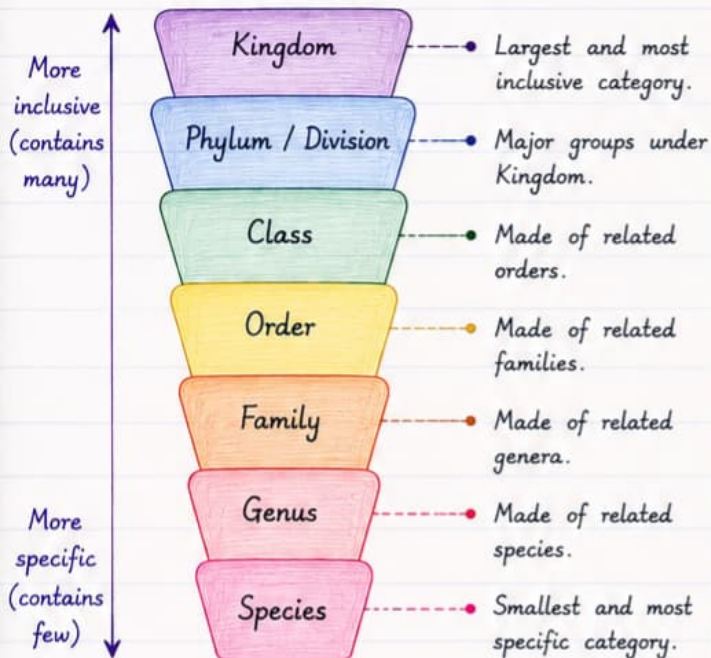


3. Taxonomic Categories

Taxonomic categories (or ranks) are a series of **hierarchical groups** used to classify organisms based on their **similarities** and **differences**.



Hierarchy of Taxonomic Categories



Example : Human Beings

Rank	Name	Explanation
Kingdom	Animalia	Animals (multicellular, eukaryotic, heterotrophic)
Phylum	Chordata	Animals having notochord at some stage of life.
Class	Mammalia	Mammals (have hair, mammary glands, warm-blooded).
Order	Primates	Primates (grasping limbs, forward facing eyes, intelligent).
Family	Hominidae	Great apes including humans.
Genus	Homo	Genus that includes modern humans.
Species	Homo sapiens	Modern human being.

Remember with Mnemonic

"King Philip Came Over For Good Soup."

K P C O F G S

Points to Remember

- ★ Taxonomic categories are based on natural relationships.
- ★ As we move from Kingdom to Species, similarity increases and number of organisms decreases.
- ★ Scientific names are universal and help avoid confusion caused by different local names.



This system was given by **Carolus Linnaeus**.

Example : Mango (Mangifera indica)

Kingdom → Plantae
 Division → Angiosperms
 Class → Dicotyledonae
 Order → Sapindales
 Family → Anacardiaceae
 Genus → Mangifera
 Species → Mangifera indica



Why Species is the Fundamental Unit?

- ◆ Similar organisms can interbreed and produce fertile offspring.
- ◆ Species represents a group that can evolve independently.
- ◆ It is the basic unit of classification.



In Short

- ☆ Taxonomic categories help us **organize** the diversity of life.
- ☆ They make the study of living beings **systematic** and easier.
- ☆ Nomenclature gives every organism a **unique scientific identity**.



4. Taxonomical Aids

Taxonomical aids are special tools and resources used by scientists to **identify**, **classify** and **study** organisms accurately.

Common Taxonomical Aids

1. Herbarium

- A herbarium is a **storehouse** of dried and pressed plant specimens.
- Each specimen is properly **labeled** with its name, habitat, date and other information.
- It is very useful in **identifying** plants.



Example

Specimens of different types of grasses, mango leaves, neem leaves, etc., are kept in a herbarium.

2. Museum



- A museum is a place where **preserved specimens** of animals, plants and other organisms are kept for reference.
- They help in the study of rare, extinct or otherwise unavailable organisms.
- They are of **two types** – **Botanical** and **Zoological**.

Example

Skeletons of animals, preserved birds, insects in resin blocks, shells, etc., are kept in museums.

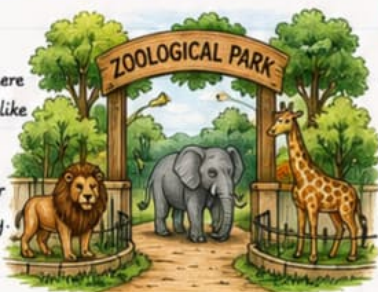
Types

- Botanical Museum** – plants, fruits, seeds, wood, etc.
- Zoological Museum** – animals, birds, insects, fishes, etc.

3. Zoological Park / Botanical Garden

(A) Zoological Park

- A zoological park is a place where **live animals** are kept in natural-like conditions.
- It helps in the study of their **behavior, habits** and life history.



(B) Botanical Garden

- A botanical garden is a place where **many species** of plants are grown for study and conservation.
- It helps in the **identification** and cultivation of plants.



4. Key

- A key is a **set of statements (clues)** used to identify an organism.
- It contains two or more **contrasting** statements called **couplets**.
- By choosing the correct statement in each step, we can reach the name of the organism.

Example of a Simple Key (for some leaves)

- 1a. Leaf simple go to 2
- 1b. Leaf compound Neem
 - 2a. Leaf margin smooth Mango
 - 2b. Leaf margin serrated Hibiscus



5. Type Specimen

- A type specimen is a **single specimen** on the basis of which the description and name of a new species is based.
- It is preserved in museums or herbaria.
- It acts as a **standard** for identifying that species in the future.

Example

The type specimen of *Mangifera indica* (mango) is preserved in a herbarium.



☆ Key Points to Remember ☆

- Taxonomical aids help in accurate identification and classification of organisms.
- They provide reference for future research and conservation.
- They make the process of taxonomy systematic, reliable and universal.

★ Without these aids, the study of vast diversity of life would be **very difficult** and full of **errors**.

